PAPER 1 Jun 2023

1 An experiment that produces a poisonous gas should be conducted in a

A classroom with closed windows.
B storeroom away from pupils.
C laboratory with work benches.
D well ventilated open space.

2 The list shows fire drill steps arranged in random order.

1. exit building in a calm and orderly manner
2. assembly at a safe place
3. sound an alarm bell
4. take a roll call

What is the correct order of the steps?

A $2 \longrightarrow 1 \longrightarrow 3 \longrightarrow 4$
B $1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4$
C $3 \longrightarrow 1 \longrightarrow 2 \longrightarrow 4$
D $4 \longrightarrow 1 \longrightarrow 3 \longrightarrow 2$

3 Which one is a function of lipids in living organism?

A formation of antibodies
B source of water
C regulation of pH
D clotting of blood

4 A substance consists of the elements carbon, hydrogen, oxygen, nitrogen and sulphur.

The substance is a

A lipid.
B protein.
C nucleic acid.
D carbohydrate.

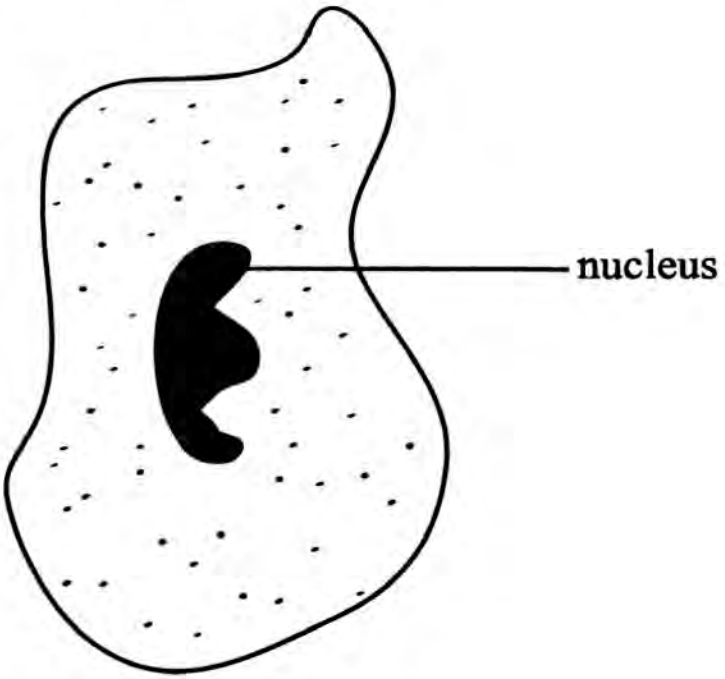
5 Cell walls are responsible for

A cell division.
B generating energy.
C structural support.
D trapping sunlight.

6 What is the likely effect of removing mitochondria from a plant cell?

A plasmolysis stops
B metabolism stops
C osmosis stops
D diffusion stops

7 The diagram shows an animal cell.



The cell is specialised for

A absorbing water.
B engulfing bacteria.
C transmitting impulses.
D transporting oxygen.

8 A biological washing powder was added to water at 60 °C.

The enzymes in the powder were

A activated.
B inactivated.
C denatured.
D saturated.

9 What are the subunits of enzymes?

A glucose
B amino acids
C fatty acids
D glycerol

10 Which biological molecule determines the structure of insulin?

A protein
B lipid
C DNA
D vitamins

11 In plants, phosphorus is needed for

A chlorophyll formation.
B ATP formation.
C fruit formation.
D flower formation.

12 Which one is a characteristic of seeds dispersed by water?

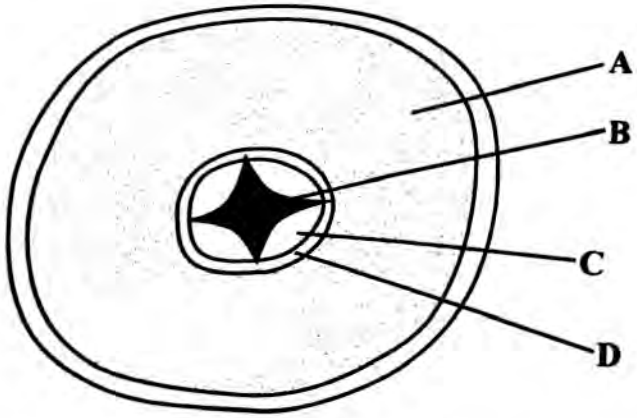
A seeds with wing like parts
B seeds with fine hairs
C seeds with fibrous coat
D seeds with hooks

13 Plants may wilt due to low

A humidity.
B temperature.
C wind speed.
D light intensity.

14 The diagram shows a transverse section of a dicotyledonous root.

Which tissue, **A**, **B**, **C** or **D**, is the xylem?



15 The concentration of carbon dioxide in the air around a plant is measured over a 24 hour period starting at midnight.

The lowest concentration of carbon dioxide will be measured at

A midnight when photosynthesis is lowest.
B sunrise when respiration rate reduces.
C midday when light intensity is brightest.
D sunset when temperatures start dropping.

16 The raw materials for photosynthesis are

A water and oxygen.
B water and carbon dioxide.
C glucose and oxygen.
D glucose and carbon dioxide.

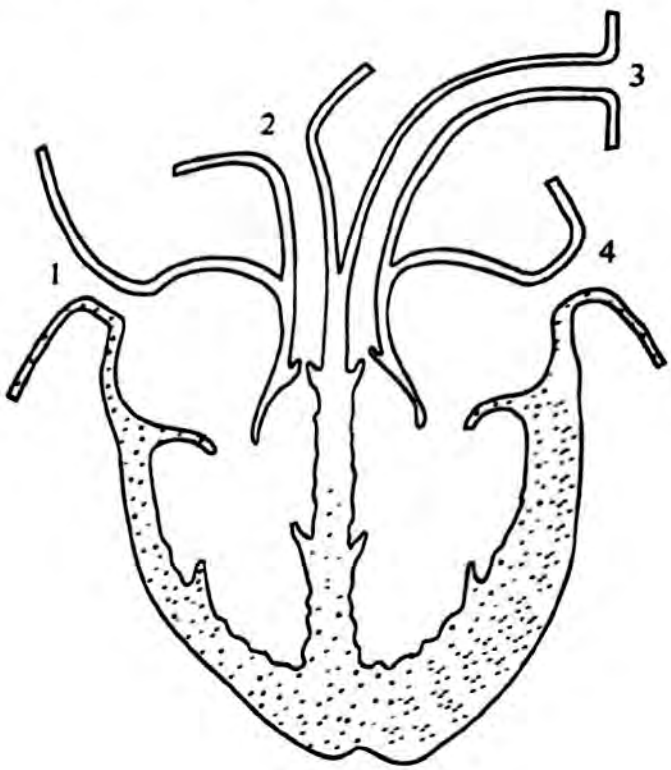
17 Which component of tobacco smoke paralyses cilia?

A tar
B nicotine
C carbon monoxide
D carbon dioxide

18 In the human kidney, the highest concentration of urea is in the

A artery.
B cortex.
C medulla.
D ureter.

19 The diagram shows the structure of the mammalian heart.

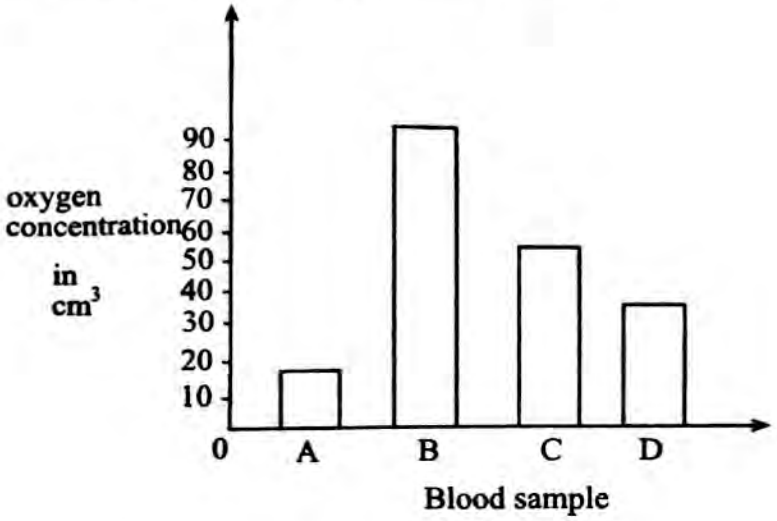


Which pair of blood vessels carry oxygenated blood?

A 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

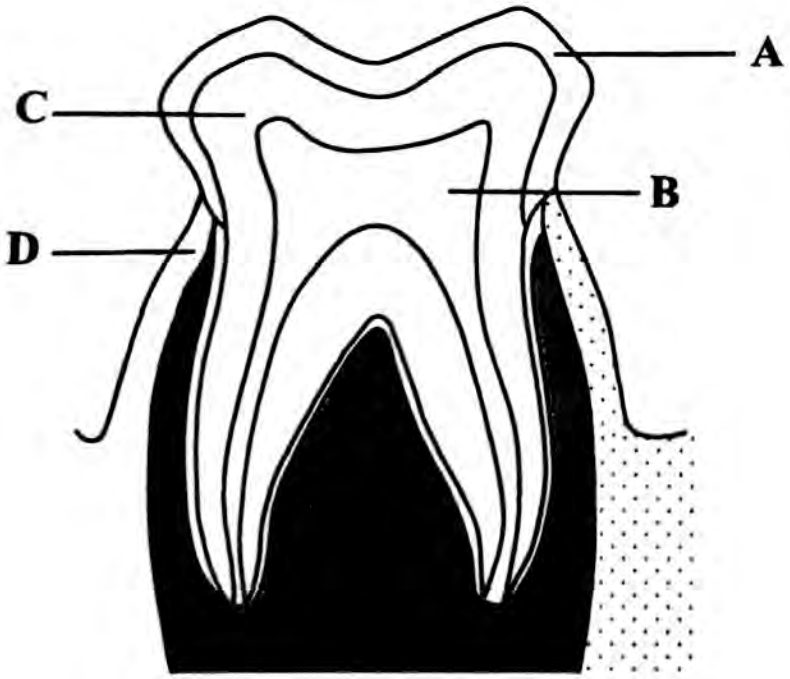
20 The bar graph shows the concentration of oxygen in blood sampled at four different places in the circulatory system of a mammal.

Which blood sample was taken from a pulmonary vein?

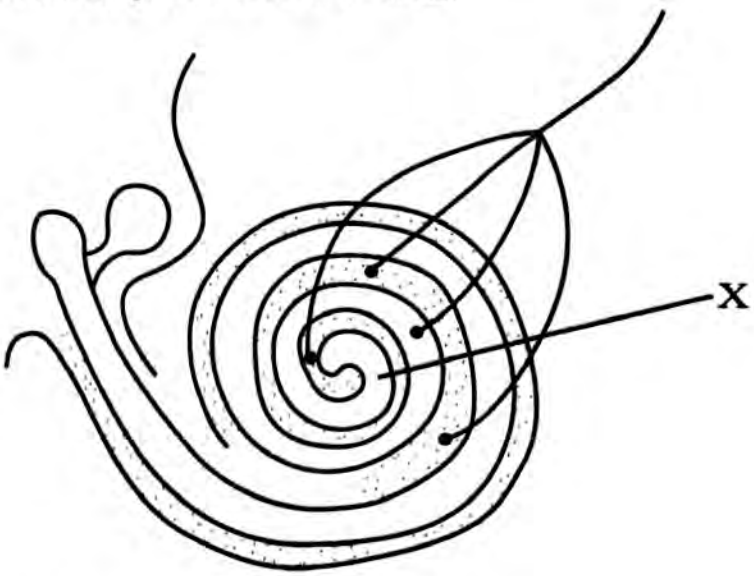


21 The diagram shows the structure of a tooth.

Which part, **A**, **B**, **C** or **D**, contains nerves and blood vessels.



22 The diagram shows a part found in the inner ear.



What is the role of the part labelled **X**?

A responds to air vibrations by moving backwards and forwards
B focuses on near and distant objects
C responds to high and low frequency sounds
D responds to light and sends impulses to the brain

23 Assimilation is

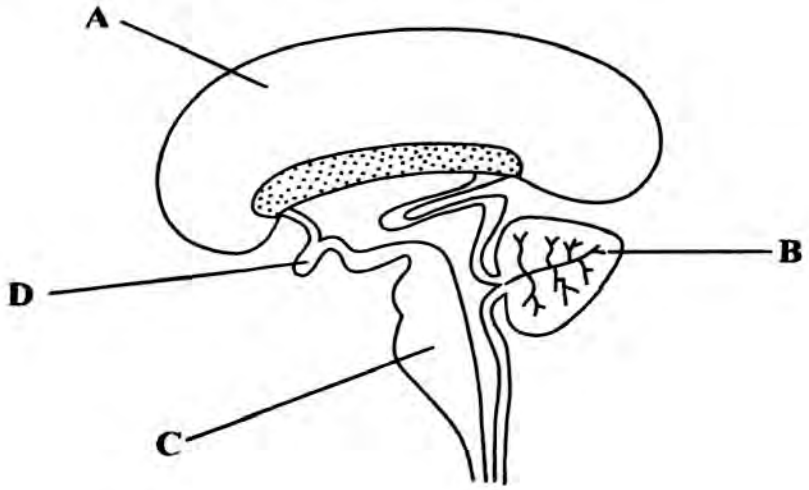
A taking food into the digestive system.
B muscular churning of food in the stomach.
C the absorption of digested food molecules from the ileum.
D making use of absorbed food molecules in metabolic processes.

24 The role of aerobic respiration is to

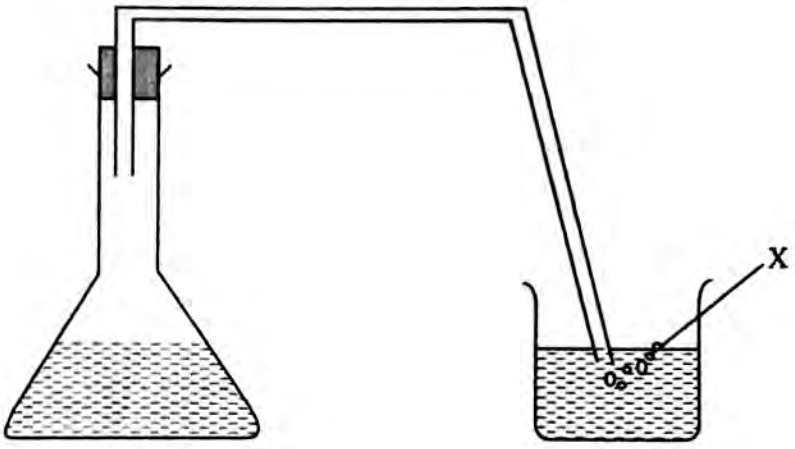
A breakdown ATP.
B produce carbon dioxide.
C produce alcohol.
D synthesize ATP.

25 The diagram shows a section through the human brain.

Which part, **A**, **B**, **C** or **D**, maintains balance?



26 The diagram shows fermentation of sugar by yeast.



yeast, sugar and water

What gas is X?

A oxygen
B carbon monoxide
C carbon dioxide
D nitrogen

27 Viruses can be viewed using

A light microscopes.
B electron microscopes.
C bio viewers.
D hand lens.

28 Which characteristic is common to **all** microorganisms?

A they cause diseases
B they are decomposers
C they are unicellular
D they contain genetic material

29 Which condition is **not** a result of gene mutation?

A albinism
B Down's syndrome
C haemophilia
D sickle cell anaemia

30 Which feature is an example of continuous variation?

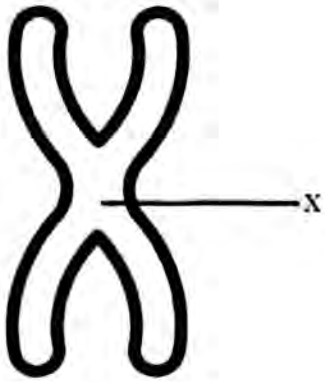
A intelligence
B tongue rolling
C earlobes
D sex

31 A man with a recessive sex linked allele is married to a normal woman. What is the probability of the allele passing on to his daughter?

A 25%
B 50%
C 75%
D 100%

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32 The diagram shows a chromosome.



The part labelled X is called the

A nucleolus.
B chromatid.
C centromere.
D allele.

33 Complete dominance is when

A different alleles have an equal effect on the phenotype of the offspring.
B an allele expresses in the phenotype even in the presence of an alternative.
C contrasting alleles are combined in an individual organism.
D a gene has more than two contrasting alleles.

34 A multi-cellular organism that has hyphae belongs to kingdom

A Animalia.
B Fungi.
C Plantae.
D Protista.

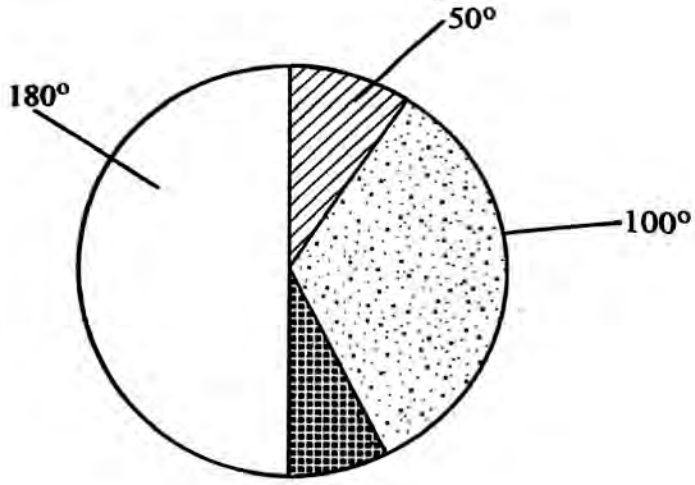
35 Which one leads to degradation of soil?

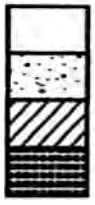
A mulching garden beds
B practicing mixed cropping
C adding manure to soil
D ploughing along slopes

36 Which human activity is a way of managing ecosystems?

A livestock rearing
B crop farming
C market gardening
D planting trees

37 The pie-chart shows components of a 50 g soil sample.





water
air content
mineral particles
organic matter

What is the mass of organic matter?

A 4.2 g
B 8.3 g
C 30 g
D 42 g

38 Liver and spleen enlargement are symptoms of

A bilharzia.
B typhoid.
C malaria.
D cholera.

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39 Which disease is associated with excessive alcohol consumption?

A kidney failure
B liver cirrhosis
C lung cancer
D tuberculosis

40 Which one is a method of preventing malaria?

A avoiding overcrowded conditions
B providing access to clean water
C provision of a balanced diet
D introducing fish into ponds

PAPER 2 June 2023

Section A

*Answer **all** questions in the spaces provided on the question paper.*

1 **(a)** **Fig.1.1** shows the structure of a palisade cell.

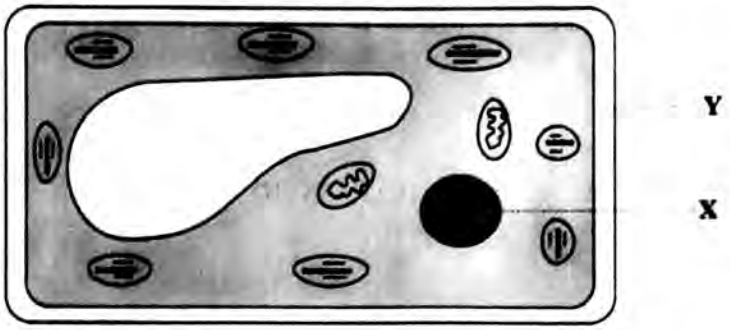


Fig. 1.1

(i) On **Fig. 1.1** label with the letter

1. A, where photosynthesis occurs,
2. B, where aerobic respiration occurs.

[2]

(ii) Name the chemical of life found in part:

X _____

Y _____

[2]

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(b) **Fig. 1.2** shows the structure of a cell involved in human reproduction.

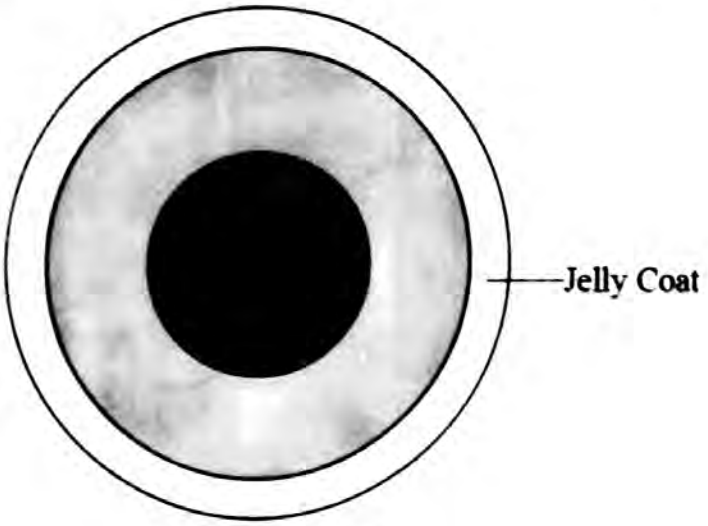


Fig. 1.2

(i) Identify the cell in **Fig. 1.2**.

_____ [1]

(ii) Give **one** structural difference between the cell in **Fig. 1.1** and the cell in **Fig. 1.2**.

_____ [1]

(iii) Relate the structure of the cell in **Fig. 1.2** to its function.

_____

_____

_____

_____ [4]

[Total: 10]

(a) **Fig. 2.1** is a graph showing the effect of temperature on the rate of an enzyme controlled reaction.

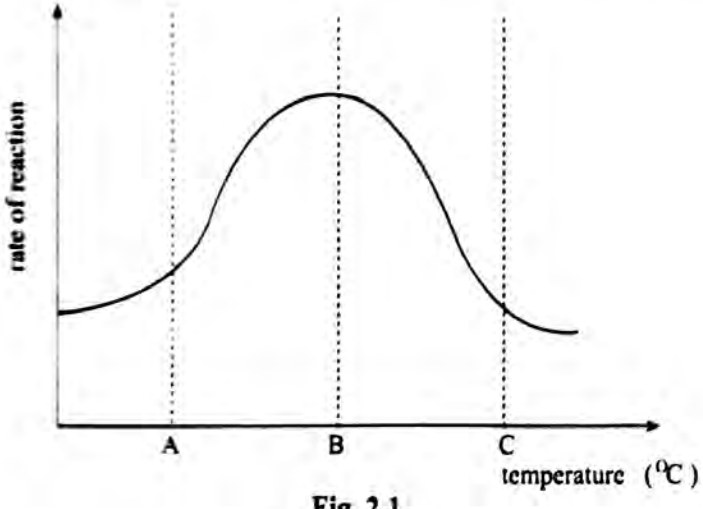


Fig. 2.1

(i) Describe the shape of the graph.

_____

_____

_____

_____ [2]

(ii) State the term that describes the temperature at B.

_____ [1]

(iii) Explain the shape of the graph between A and B.

_____

_____

_____

_____

_____

_____ [3]

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(iv) Explain how the temperature at C illustrates the protein nature of enzymes.

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_____

_____

_____

_____ [2]

(b) State any **two** other properties of enzymes.

1. _____

2. _____ [2]

[Total: 10]

3 (a) **Fig. 3.1** shows a cross section of a dicotyledonous root.

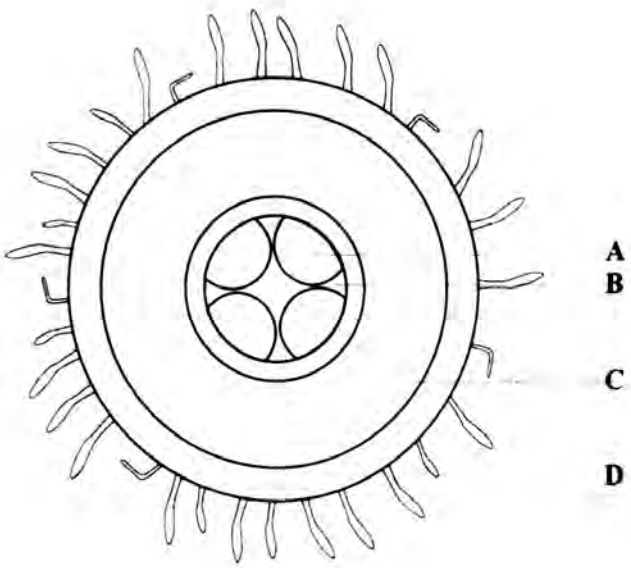


Fig. 3.1

(i) Name the part labelled:

A _____

B _____

C _____

D _____ [4]

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(ii) Describe the functions of

B. _____

_____

D. _____

_____ [2]

(b) Explain the effects of wilting on a plant.

_____

_____

_____

_____ [4]

[Total: 10]

4 (a) State the function, in the mammalian ear, of the

(i) pinna,

_____

_____ [2]

(ii) auditory canal,

_____

_____ [2]

(iii) cochlea,

_____

_____ [1]

(iv) semi circular canals.

_____

_____ [1]

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(b) **Suggest two effects of a very loud explosion to the ear**

1 _____

_____

2. _____

_____ **[2]**

(c) **Explain how a person with hereditary deafness can be assisted.**

_____

_____

_____

_____ **[2]**

[Total: 10]

5 **(a)** **(i)** Name the substance found in the core of a virus.

_____ **[1]**

(ii) State any **two** diseases that are caused by a virus.

1. _____

2. _____ **[2]**

(b) Suggest why viruses are **not** considered as **living** organisms.

_____

_____

_____

_____

_____

_____

_____ **[4]**

(c) State any **three** structural features of bacteria.

1. _____

2. _____

3. _____ [3]

[Total: 10]

6 (a) Define the term *co-dominance*.

_____

_____

_____ [2]

(b) State any **two** diseases caused by a gene mutation.

1. _____

2. _____

[2]

(c) Name any **two** factors that increase the rate of mutation.

_____

_____ [2]

A man heterozygous for blood group **B** and a woman of blood group **AB** produced a child.

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Using a genetic diagram, show the probability of the child having blood group **O**.

Section B

*Answer any **four** questions on the separate answer paper provided*

7 **(a)** Describe the safety rules in the handling of microorganisms in a laboratory [6]

(b) Explain why learners and staff at a school should perform fire drills [4]

8 **(a)** Relate the property of water to its function as a

1 solvent.

2 reagent

[6]

(b) State any **four** uses of lipids in the body [4]

9 **(a)** Describe how tooth decay occurs [4]

(b) Explain how the use of toothpaste prevents tooth decay [6]

10 **(a)** Describe measures taken to conserve an endangered animal species [6]

(b) Explain how industrial effluent is a threat to biodiversity [4]

11 **(a)** Describe the role of microorganisms in the nitrogen cycle [6]

(b) Explain the causes of low plant species diversity in artificial ecosystems [4]

12 **(a)** Describe any **three** causes of diseases [6]

(b) (i) Differentiate between an infectious disease and non infectious disease [2]

(ii) State one example of

1 an infectious disease.

2 a non infectious disease

[2]

Candidate Name	Centre Number	Candidate Number

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

BIOLOGY 4025/3

PAPER 3 Practical Test

JUNE 2023 SESSION 1 hour 30 minutes

Candidates answer on the question paper

Additional materials:

As listed in Instructions to Supervisors
Electronic calculator
Pencil (B or HB is recommended)
Soft clean eraser
Ruler (cm/mm)

TIME 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Use a sharp pencil for your drawings.

Coloured pencils and crayons should **not** be used.

FOR EXAMINER'S USE	
1	
2	
TOTAL	

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

This question paper consists of 6 printed pages and 2 blank pages.

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1 (a) You are required to compare the amount of sugar in three different food extracts. You are provided with the three food extracts, **A**, **B** and **C**.
Carry out the Benedicts test to compare the amount of sugar in the food extracts.
Proceed as follows:
Label three test tubes **A**, **B** and **C**.
Using a syringe, put 2 cm^3 of extract **A** into test tube **A**.
Put 2 cm^3 of extract **B** into test tube **B**.
Put 2 cm^3 of extract **C** into test tube **C**.
Add 2 cm^3 of Benedicts solution into each of the test tubes **A**, **B** and **C**.
Shake the test tubes to mix the contents.
Place the test tubes into a boiling water bath for 2 minutes.
After the 2 minutes, observe the contents of the test tubes.

(i) Record and explain the observations made in **Table 1.1**

Table 1.1

Food extract	Observation	Explanation
A		
B		
C		

[6]

(ii) Suggest how the procedure may be improved to obtain more accurate estimates of the amount of sugar in the food extracts.

_____

_____

_____

_____

_____

_____ [4]

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(iii) State and explain, which food extract would be recommended for an active athlete.

food extract _____

explanation _____

_____

_____ [3]

(b) Draw a labelled diagram to show the set up of apparatus used in determining the energy content of a food sample.

[7]
[Total: 20]

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2 You are required to examine the external structure and the internal structure of two different types of seeds.

You are provided with soaked seed **A** and soaked seed **B**, a hand lens and a scalpel.

(a) **(i)** Examine the external structure of seed **A** and seed **B** in turn using the hand lens.

Draw labelled a diagram to show the external structure of

seed **A**,

seed **B**.

[6]

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(ii) Identify seed

A, _____

B. _____

[2]

(b) (i) Using the scalpel blade, split seed **A** longitudinally into two halves.
Examine the internal structure of seed **A** using the hand lens.

Draw a labelled diagram to show the internal structure of seed **A**.

[4]

(ii) Cut seed **B** longitudinally using the scalpel blade.
Examine the internal structure of seed **B** using the hand lens.

Draw a labelled diagram to show the internal structure of seed **B**.

[4]

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(c) **(i)** **Identify two visible structural differences between seed A and seed B.**

1. _____

2. _____

[2]

(ii) **State the function of any visible part of seed**

A, _____

_____

_____

B. _____

_____

_____

[2]

[Total: 20]

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